

Do Duc Anh Nguyen

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Education

- 2020–2022 **Master of Security Information**, University of Limoges, France
GPA M1: 14.877/20 - Rank 4/20
GPA M2: 14.836/20 - Rank 5/26
- 2015–2019 **Bachelor of Engineering, Computer Science**, University of Technology (HCMUT), Vietnam
GPA: 7.94/10 - HONORS PROGRAM
- **Traffic Light Control System (Thesis):** Using Deep Reinforcement Learning to implement a model to control the traffic light aiming to reduce the waiting time of vehicles. Built a simulation system, analyzed different models and training strategies
- Score: 9.6/10

Working Experience

- 2023–2026 **Doctoral Candidate in Information Security**, IMT Atlantique, Rennes, France
- **Research on Opportunistic Reactions to Malware Propagation**
 1. Proposed a traffic segmentation methodology to derive malicious behavior from observed attack traffic, applying it specifically to the WannaCry use case
 2. Modeled network behavior for self-propagating malware to appropriately design and evaluate theoretical countermeasure performance
 3. Designed signaling methodologies that integrate security microservices by opportunistically leveraging existing traffic for information exchange and rapid response
 4. Applied distributed, self-stabilizing vertex cover algorithms to optimize the mathematical placement of Intrusion Detection System (IDS) microservices
 - **Lab teaching in Network Security - 36 hours**
Prepare and evaluate labs for ~ 20 students per lab
 - **Academic Service**
A reviewer for a paper at the IEEE/IFIP NOMS conference, 2025
- 2022 **Master Intern**, IMT Atlantique, Rennes, France
- **Research on Intent-Based Networking (IBN)**
Select the use case of existing IBN framework, Interface to Network Security Function (I2NSF), to evaluate our policy conflict detection mechanism performance
- 2018–2020 **Backend Developer**, Zalo Group, VNG Corporation, HCMC, Vietnam
- **Major servers building**
Developed and maintained new services on central servers with association of database NoSQL database to handle product features of Zalo application
 - **Code Optimization & Congestion Management**
Handled a vast number of requests of more than 100 million users per day by permanently testing and optimizing the system. Managed troubleshoot server congestion in rush hours
- 2018–2020 **Research Member**, ML4U Research Group, HCMUT, Vietnam
- **Experience sharing**
Participated in weekly seminars about Machine Learning coordinated by lecturers at HCMUT. Organized a seminar every two weeks for presentation

Publications

- Do Duc Anh Nguyen, Pierre Alain, Fabien Autrel, Ahmed Bouabdallah, Jérôme François, and Guillaume Doyen. “How Fast Does Malware Leveraging EternalBlue Propagate? The case of WannaCry and NotPetya” SecSoft Workshop, 2024
- Do Duc Anh Nguyen, Pierre Alain, Fabien Autrel, Ahmed Bouabdallah, and Jérôme François. “Intent-based Attack Mitigation Through Opportunistic Synchronization of Microservices” NetSoft PhD Symposium, 2024
- Do Duc Anh Nguyen, Fabien Autrel, Ahmed Bouabdallah, and Guillaume Doyen. “A Robust Approach for the Detection and Prevention of Conflicts in I2NSF Security Policies.” NOMS 2023-2023 IEEE/IFIP Network Operations and Management Symposium. IEEE, 2023

Achivements & Scholarships

- **Full Studying Overseas Scholarship**, Vingroup, 2020
- **Scholarships** for Outstanding Student, HCMUT, 2019
- **Scholarships** for Talent Students in Honors Program, HCMUT 2016, 2017, 2018
- **Highest Score Students** (Top 0.1%/300.000) in National High School Graduation Examination, 2015

Skills

- Programming Languages: Java, C/C++, Python, Ocaml
- Languages: English (advanced), French (intermediate)
- Technology experience: GNS3, OMNet++, MirageOS

Projects

- **Opportunistic Synchronization Mechanism**, 2025
Implemented a fast-reaction mechanism leveraging existing traffic to deliver information between security functions (microservices) across both emulation (GNS3) and simulation (OMNET++) environments
- **Malware Propagation Testbed**, 2024
Built and configured an emulation testbed to dynamically analyze behavior and measure the real-world propagation speeds of WannaCry and NotPetya malware
- **Policy Conflict Detection and Resolution for I2NSF Framework**, 2023
Implemented a new conflict policy detection and resolution mechanism for the I2NSF framework proposed by the IETF. This user-level mechanism provides meaningful feedback for conflict resolution

Social activities

- **Treasurer, ADER (Association de Doctorants de l'ENST de Bretagne, campus de Rennes):**
Managed the budget for doctoral after-work events (2023–2025)
- **Green Summer Voluntary Campaign:** Participated in building roads in a disadvantaged village in Ben Tre Province (2017)
- **Mid-autumn Festival:** Organized a Moon Festival for an elementary school with food and handmade lantern in Ben Tre province (2016, 2017)
- **The House of Love::** Contributed food, handmade toys to orphans and helped them to clean their home in Binh Duong province. (2015)

References

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| ○ Prof. Guillaume Doyen
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